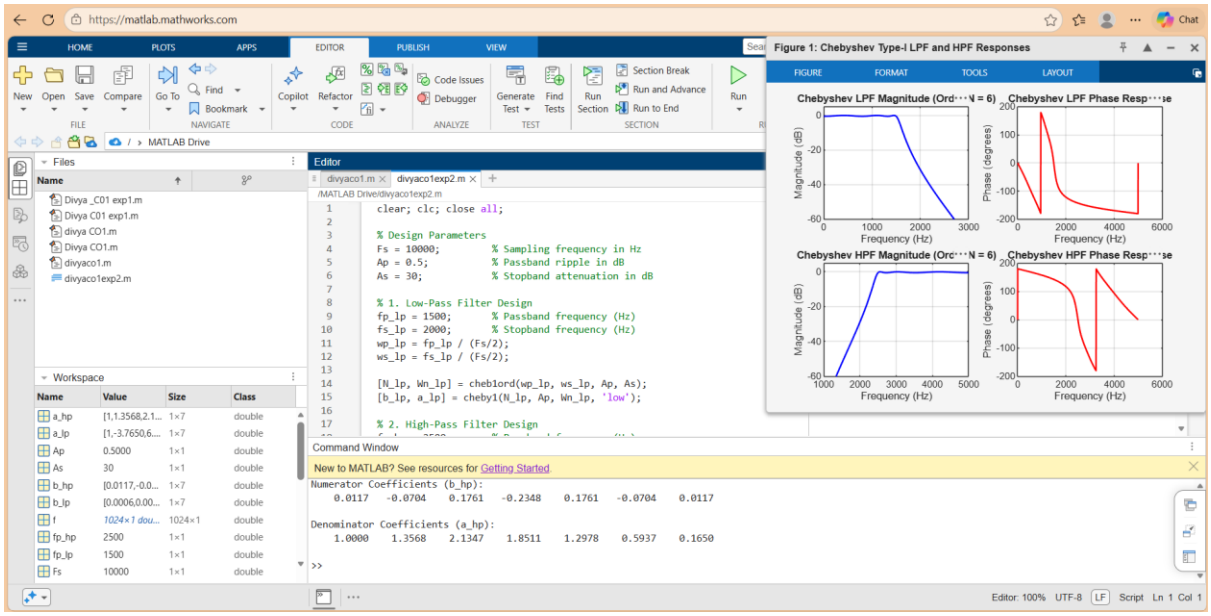

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Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
Editor
divyaco1.m
divyaco1exp2.m
/MATLAB Drive/divyaco1exp2.m
1 clear; clc; close all;
2
3 % Design Parameters
4 Fs = 10000; % Sampling frequency in Hz
5 Ap = 0.5; % Passband ripple in dB
6 As = 30; % Stopband attenuation in dB
7
8 % 1. Low-Pass Filter Design
9 fp_lp = 1500; % Passband frequency (Hz)
10 fs_lp = 2000; % Stopband frequency (Hz)
11 wp_lp = fp_lp / (Fs/2);
12 ws_lp = fs_lp / (Fs/2);
13
14 [N_lp, Wn_lp] = cheb1ord(wp_lp, ws_lp, Ap, As);
15 [b_lp, a_lp] = cheby1(N_lp, Ap, Wn_lp, 'low');
16
17 % 2. High-Pass Filter Design
Workspace
Name
Value
Size
Class
a_hp [1,1.3568,2.1... 1×7 double
a_lp [1,-3.7650,6.... 1×7 double
Ap 0.5000 1×1 double
As 30 1×1 double
b_hp [0.0117,-0.0... 1×7 double
b_lp [0.0006,0.00... 1×7 double
f 1024×1 dou... 1024×1 double
fp_hp 2500 1×1 double
fp_lp 1500 1×1 double
Fs 10000 1×1 double
Command Window
New to MATLAB? See resources for Getting Started.
Numerator Coefficients (b_hp):
0.0117 -0.0704 0.1761 -0.2348 0.1761 -0.0704 0.0117
Denominator Coefficients (a_hp):
1.0000 1.3568 2.1347 1.8511 1.2978 0.5937 0.1650
>>
Editor: 100% UTF-8 LF Script Ln 1 Col 1
Figure 1: Chebyshev Type-I LPF and HPF Responses
FIGURE
FORMAT
TOOLS
LAYOUT
Chebyshev LPF Magnitude (Ord···N = 6)
Chebyshev LPF Phase Resp···se
Chebyshev HPF Magnitude (Ord···N = 6)
Chebyshev HPF Phase Resp···se
Magnitude (dB)
Phase (degrees)
Frequency (Hz)